Daily Drilling Report
SPD19B-03
8/26/2014

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
30.0	12 1/4"	175.00	72.83	31.70	92.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
13 3/8" Shoe, 1,487.00mKB	235,715.60	9,653.92	6,992,375.62		24.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,224.00	2,377.00	2,254.95	153.00	18.67	8.2

Operations at Report Time
Drilling @ 2421m.

Operations Summary
Dir. drilling 12 1/4" hole section f/ 2224m to 2377m.

Operations next 24hrs
Cont' drilling 12 1/4" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Jabbar Mohammadi	Night Drilling Sup.
Hamed Daneshkhah / B.Khalilmoghadam	Drilling Engineer
Ali Sahebnouri	Geologist
R. Kamalpour	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	8/20/2014	9/11/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	8/16/2014	9/16/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	4/16/2014	6/16/2014
Pit Drill	8/25/2014	10/10/2014
Trip Drill	8/21/2014	10/11/2014

STOP Cards

Number of Reports: 40

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00		12 1/4"	D	03	P	Directional drilling 12 1/4" hole section in rotation & slide mode f/ 2224m to 2377m w/ 800-850 gpm/ 1800-1920 psi/ 32-37 kips/ 45 rpm/ 11-24 kft-lbs. Rotation Interval Slide Interval 2224m-2292m 2292m-2302m 2302m-2377m 2112m-2146m >> Increase MW f/ 10.2 ppg to 10.8 ppg before entering "Upper Dashtak" member and increase to 11.2 ppg while entering "Evaporite" member. ** Drilling with NL/NG. ** Pump 20 bbl Hi-Vis pill when Std down. ** B/R and R/D once each Std drilled . ** Take survey before connection.
00:00	00:00		12 1/4"				Aug 27, 2014 00:00 - 06:00 Cont' directional drilling 12 1/4" hole section in rotation mode f/ 2377m to 2421m w/ 800 gpm/ 1900 psi/ 37 kips/ 45 rpm/ 11-23 kft-lbs. >> Last sample @ 2410m: Arg Lst (50 %) + Clst (30%) + Dol (10%) + Anhy (10%).

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,377.00	KCl-Polymer Mud	11.20	52	22.0	21.000	4.7
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
9.5	0.100	0.800	0.400			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
3	5	2.0	83.0	15.0	44,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
5.000	8.000	11.000	2.4/97.6	54.0	1894.0	1890.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
4202.0	134.0	0.0	0.01		6.060889	0.569

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PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
0.50	7.	200	5.0	999.0	34.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Hani	Supply Boat	8/26/2014 15:55		Delivered FO-120 MT. Dep for Aban-8 at 2225 Hrs.
Marina-1	Supply Boat	8/16/2014 15:15		At Sahar-2 Location.
C-Mariner	Supply Boat	8/25/2014 21:45		Delivered cargo. Received FO-30 MT f/ DP3. Dep for Aban 8 at 1635 Hrs.
Moje Darya	Supply Boat	8/27/2014 00:45		Delivering Cement in progress.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Barite Bulk	30.0	249.0	207.0
Caustic Soda 2014	16.0	244.0	149.0
CaCo3-C 25Kg	30.0	40.0	10.0
MI-Cide	1.0	15.0	9.0
Defoamer	1.0	8.0	6.0
Pac-LV	2.0	262.0	36.0
Soda Ash 25 Kg	22.0	350.0	165.0
Starch	80.0	680.0	260.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Cement G	0.0	0.0	57.0
Drill Water	10.0	0.0	343.0
Diesel Fuel	13.0	120.0	205.0
Potable Water	20.0	20.0	110.0
Barite BB	0.0	0.0	0.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0
Barite Bulk	30.0	0.0	207.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
8	12 1/4	BEST	M1385	228378	PDC	M432	12/12/12/12/13/13/13
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.95	87.10	679,343.40	1,495.00	2,377.00	882.00	10.1	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
750	880	1,050.0	1,920.0	9	37	97	179
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
113.83	51.19	58.77	41.23	380	339	243	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,224.00	2,292.00	5.55	23,000.0	800	45	1,850.0	35	330
2,292.00	2,302.00	3.57		850		1,920.0	25	330
2,302.00	2,377.00	9.55	23,000.0	800	45	1,900.0	37	339

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.250	27.83	226.57
7" Drilling Jar (COR-3022-AD)	7	3.000	5.85	198.74
16 jts 5 7/8" HWDP	5 7/8	4.250	149.23	192.89
X/O	7	2.810	0.79	43.66
1 jt 8 1/4" DC	8 1/4	2.810	9.46	42.87
8 1/2" Float Sub	8 1/2		0.79	33.41
8" NMDC	8	3.500	9.44	32.62
8" TM - Sub	8		2.03	23.18
8" NMDC	8	3.500	9.46	21.15
12" S. Stabilizer (COR3059AD)	11 7/8	3.125	2.19	11.69
9 5/8" Motor (AKO: 1.00° , Sleeve: 12 1/8")	9 5/8		9.14	9.50
12 1/4" PDC bit	12 1/4		0.36	0.36

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)
2	2,259.00	14.00	20	150.0	97	114
2	2,259.00	14.00	30	205.0	97	170
2	2,259.00	14.00	40	280.0	97	227
3	2,259.00	14.00	20	150.0	97	114
3	2,259.00	14.00	30	205.0	97	170
3	2,259.00	14.00	40	290.0	97	227

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7

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Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,212.28	245.70	22.60	2,102.55	-269.12	-575.47	635.29	0.42	-0.31
2,241.35	245.90	22.60	2,129.38	-273.70	-585.66	646.46	0.08	0.00
2,270.15	245.90	22.20	2,156.01	-278.18	-595.68	657.43	0.42	-0.42
2,299.45	246.00	22.40	2,183.12	-282.71	-605.83	668.55	0.21	0.20
2,328.29	246.20	22.30	2,209.79	-287.15	-615.86	679.52	0.13	-0.10
2,357.13	246.40	22.00	2,236.50	-291.53	-625.82	690.39	0.32	-0.31
2,386.11	246.70	21.50	2,263.42	-295.80	-635.67	701.12	0.53	-0.52

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
L.Sur.Sh.		2,161.00	2,055.29
Lithiotis Bed.		2,218.50	2,108.29
Neyriz		2,240.00	2,128.14
Up Dashtak		2,266.00	2,152.17
Evaporite "B"		2,338.00	2,218.78

General Remarks

Comment

>>> Requirements: (Outstanding of MI MR#12: 2 Pallets of Pac-LV) + (60 MT Cement) + (100 MT FO) + (2 x O-Ring for 10 3/4" Pack off RT) + (MI MR#13).

>>> Sent: Nil.

>>> Received:
** Hani f/ Sharjah: (120 MT FO).